

Rythmiq

Lab 1 - Product Description

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1 Introduction

Independent cafes, coffee shops, and local restaurants depend on atmosphere as part of the customer experience. Background music is one environmental factor that can influence customer behavior, satisfaction, and time spent in a venue (Milliman, 1982; North et al., 1999). However, many small venues do not have an affordable and practical way to let customers interact with the music while preserving the atmosphere selected by the business. The Rythmiq project focuses on small hospitality venues, particularly locations with fewer than 50 seats, where proprietary music-management hardware can be too expensive and verbal song requests can interrupt employees during busy service periods.

Existing approaches do not fully address this problem for small businesses. TouchTunes and Rockbot provide business-oriented music solutions, but cost, hardware, and target-market limitations can make these systems less accessible to small venues (Rockbot, n.d.; TouchTunes, n.d.). Manual song requests require customers to interrupt staff and can produce inconsistent results when employees accept requests that do not match the intended atmosphere. Personal streaming services are also not an appropriate substitute for licensed commercial playback; Spotify's terms restrict personal-use services from commercial use (Spotify, 2023). Small venues therefore need a lower-cost, hardware-agnostic system that supports licensed commercial playback, reduces staff interruptions, and preserves the venue's control over its music.

Rythmiq, summarized by the tagline “Find Your Rhythm,” directly addresses those needs through a web-based subscription jukebox platform. Customers use their own smartphones to submit requests, while venue owners and staff establish the musical constraints that protect the business’s intended atmosphere. By avoiding proprietary hardware, automating request screening, supporting licensed commercial playback, and offering optional paid requests,

Rythmiq provides a software-centered alternative designed specifically for small local spaces. The following product description explains how these capabilities work together.

2 Product Description

Rythmiq is a software-only music request and management system that replaces the informal process of customers verbally asking employees to change or play music. It is designed to operate on internet-connected devices that a business already owns, eliminating the need for dedicated proprietary hardware. Customers access a venue-specific web portal from their smartphones to browse eligible songs, view the active queue, and submit requests. Owners and staff use a separate Business Control Center to define musical restrictions, manage the queue, and review activity. Together, these functions give customers a convenient way to participate in the venue's music while allowing the business to preserve its intended atmosphere.

The product has five primary objectives. It must reduce staff interruptions, maintain venue control over requested music, support commercially licensed playback, remain affordable for small businesses, and create an optional revenue opportunity. Rythmiq meets these objectives through controlled customer access, automatic validation, a hardware-agnostic design, integration with third-party music services, and optional paid request prioritization.

2.1 Key Product Features and Capabilities

QR Code Access provides the entry point to Rythmiq. A venue displays its unique QR code at tables, counters, or other customer areas. Scanning the code opens the venue's customer portal in a mobile browser, so the customer does not have to install an application or create a new interaction with an employee. The code links the session to the correct venue and ensures that the customer sees that venue's queue and available music options.

Song Browsing and Requests allow customers to search the available catalog and submit a selection from their own phones. Music metadata is supplied through the planned MusicBrainz integration, and the interface displays only the information needed to identify and request a song. Customers can also view the current queue before submitting a request, which helps reduce duplicates and makes the request process more predictable (MetaBrainz Foundation, n.d.).

Dynamic Vibe Moderation is performed by the Vibe Engine, Rythmiq's validation logic. Before a request enters the queue, the engine compares available song attributes with restrictions stored for the venue. A compliant request is accepted and placed in the queue, while a request that violates the configured rules is rejected. This automatic screening lets customers participate without giving them unrestricted control and prevents employees from having to evaluate every request manually.

The Business Control Center gives owners and authorized staff a real-time interface for configuring venue rules and supervising activity. Users can review and adjust the queue, select permitted genres or moods, block explicit content, define tempo limits, and view request and transaction information. Because these settings are stored as a venue profile, the business can apply consistent standards across a shift and revise them when the desired atmosphere changes.

Pay-to-Request is an optional feature that allows a customer to pay approximately \$0.99 to \$1.99 to increase a qualifying request's priority. Payment does not allow a song to bypass venue restrictions; the Vibe Engine must approve the request before it can be promoted. The planned Stripe API integration processes payment requests, while the proposed model reserves a five-percent platform share for Rythmiq (Stripe, Inc., n.d.). Standard unpaid requests remain available so participation is not limited to paying customers.

Licensed playback is planned through SoundMachine, a commercial music service intended for business environments (SoundMachine, n.d.). Separating customer requests from the licensed playback service helps Rythmiq avoid treating a personal streaming account as a commercial music source. In combination, these capabilities move song requests into a controlled digital workflow, reduce interruptions during service, and give small venues a practical alternative to a dedicated jukebox.

2.2 Major Components (Hardware/Software)

Rythmiq does not require proprietary hardware. The intended venue environment consists of an internet-connected tablet, smartphone, or computer; the venue's existing speaker system; and a reliable network connection. Customers provide their own smartphones to access the request portal. Reusing equipment that is already present lowers the cost of deployment and supports the product's goal of serving small businesses that may not be able to justify a dedicated music-management device.

The Web App User Interface is the mobile-first customer portal opened from the venue's QR code. It presents song search, queue viewing, request submission, and payment options in a browser. The Business Control Center Dashboard is the owner- and staff-facing interface. It provides controls for queue management, venue restrictions, account configuration, and analytics. Separating these interfaces limits customers to request-related actions while reserving administrative controls for authorized venue users.

The Vibe Engine is the central custom software component. It receives a request from the Web App User Interface, obtains or evaluates the associated metadata, compares the result with the venue's stored rules, and returns an approval or rejection. The Venue Database supports this process by storing venue profiles, configuration rules, queue state, account information, and

transaction records. The Local Server coordinates the current playback state and communication among the customer portal, validation logic, database, management dashboard, and licensed playback service.

Three planned third-party services supplement the custom components. MusicBrainz supplies song metadata used during search and request evaluation (MetaBrainz Foundation, n.d.). Stripe processes optional pay-to-request transactions (Stripe, Inc., n.d.). SoundMachine provides the commercially licensed music playback layer (SoundMachine, n.d.). These services perform distinct roles: MusicBrainz identifies music, Stripe handles payment requests, and SoundMachine supplies the audio played in the business. The team moved away from the Spotify API because Spotify's personal-use terms do not permit commercial playback (Spotify, 2023).

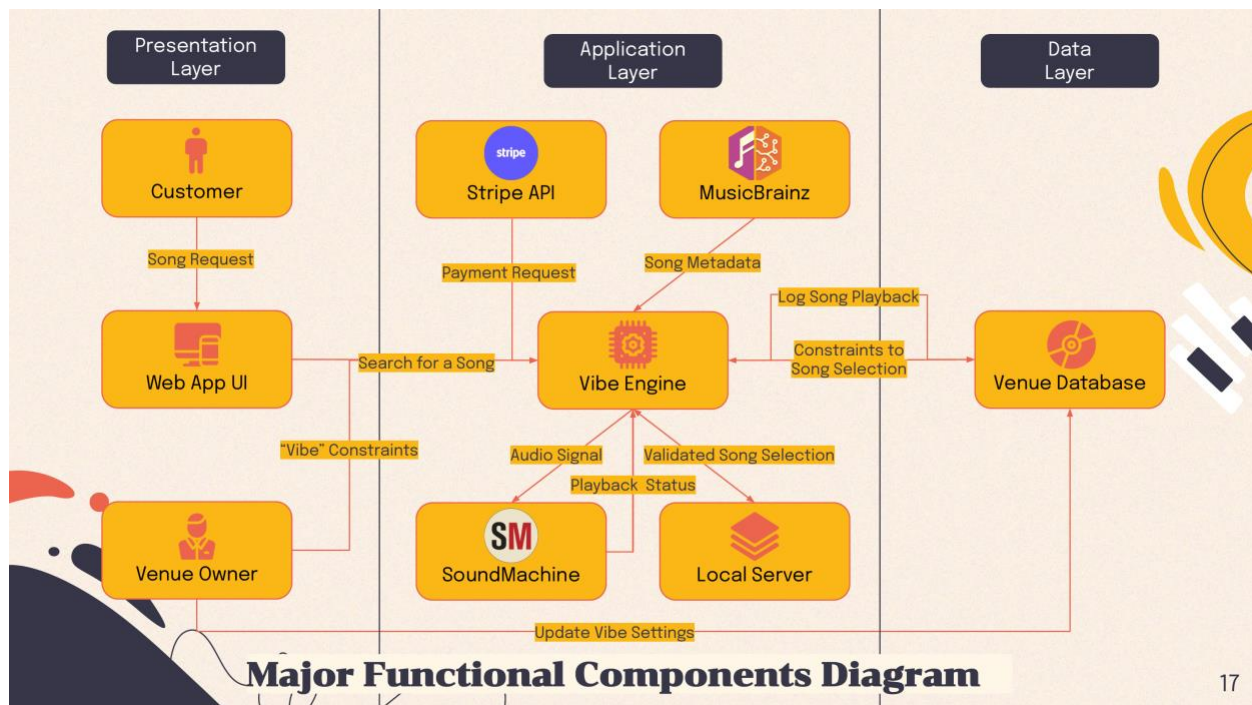


Figure 1. Rythmiq Major Functional Components Diagram

Note. The diagram represents the planned real-world product architecture.

Figure 1 organizes the planned architecture into presentation, application, and data layers. In the presentation layer, customers send song requests through the Web App User Interface, while venue owners provide vibe constraints and update venue settings. In the application layer, the Vibe Engine receives search and metadata inputs, evaluates each request, communicates payment requests through Stripe, sends validated selections to the Local Server, and coordinates playback with SoundMachine. In the data layer, the Venue Database stores constraints and song-playback records. The arrows show that customer and owner actions do not pass directly to playback; each request moves through validation and the stored venue rules before it can affect the queue. This separation preserves business control while allowing customers to interact with the music.

The planned technology stack is LAMP: Linux as the operating environment, Apache as the web server, MySQL as the database management system, and PHP for server-side application logic. A responsive mobile-first front end supports the customer and business interfaces, and a QR-code generation library creates venue-specific access links. This stack uses widely supported web technologies and does not require customers to install platform-specific software.

In the intended workflow, a customer scans the venue's QR code, searches for a song, and submits a request through the Web App User Interface. MusicBrainz returns the song metadata, and the Vibe Engine checks the request against the venue profile stored in the Venue Database. If the request satisfies the rules, the system adds it to the queue and coordinates licensed playback through the Local Server and SoundMachine; otherwise, it returns a rejection to the customer. When a customer selects paid priority, Stripe handles the payment request without bypassing validation. Staff can monitor the process or change venue settings through the

Business Control Center without handling each request individually. This division of responsibilities connects every major component to the core problem and provides the foundation for the case study in Section 3.

3 Identification of Case Study

The Rythmiq case study introduces a small beta-test group at “Brewpoint Cafe,” a representative single-location independent coffee shop with approximately 30 seats. The participants include the cafe owner, two baristas, and a small group of regular customers. The owner acts as the subscribing customer and system administrator by defining the musical atmosphere and reviewing activity. The baristas act as authorized staff members who can monitor the queue without managing each request individually. Cafe patrons act as end users who search for songs and submit requests from their own smartphones. This group represents the intended early market for Rythmiq: small hospitality venues that want customer participation without sacrificing staff time or control of the environment.

The case study exercises the prototype features most closely connected to the identified problem: QR Code Access, Song Browsing and Requests, Dynamic Vibe Moderation, the Business Control Center, queue management, and the Pay-to-Request option. These features produce a different benefit for each participant. Customers gain a direct and predictable method for influencing the music. Baristas avoid repeated verbal interruptions during service. The owner maintains an acoustic, indie, and lo-fi atmosphere by blocking explicit content, limiting tempo to 120 beats per minute, and reviewing request activity. The paid-request option also demonstrates how the venue could earn revenue from customer engagement without allowing payment to override its rules.

A typical “day in the life” demonstration begins before the morning rush. The owner opens the Business Control Center on the cafe’s existing tablet, selects the permitted genres, blocks explicit songs, and confirms the tempo limit. A barista places the venue’s QR code near the register and on customer tables. When customers arrive, they can scan the code and open the

Rythmiq web portal without downloading an application or asking an employee for assistance. The current queue and eligible search results are available from each customer's phone.

During the demonstration, a representative customer named Alex scans the code, views the queue, and searches for "Golden Hour" by JVKE. Alex submits the song, and the Vibe Engine compares its available metadata with Brewpoint Cafe's stored genre, explicit-content, and tempo restrictions. If the song satisfies every rule, the prototype accepts it and adds it to the queue; if it violates a rule, the interface rejects it without requiring a barista to intervene. Alex may then use the optional \$0.99 boost to raise the approved request's priority. The baristas continue serving customers while the owner monitors the queue and request information through the Business Control Center. This scenario demonstrates the complete prototype workflow from venue configuration and customer access through validation, queuing, staff monitoring, and licensed playback coordination.

The prototype demonstration is expected to show whether customers can complete requests without staff assistance, whether the Vibe Engine consistently applies the owner's restrictions, and whether authorized staff can observe and adjust the queue from an existing device. A successful result would show that the request reaches the correct venue, prohibited selections do not enter the queue, approved selections appear in the proper order, paid priority is applied only after validation, and the owner can review the resulting activity. These results would demonstrate how Rythmiq reduces service interruptions while preserving the cafe's intended atmosphere.

In a full-capability implementation, Rythmiq could add secure venue accounts, integrated payment processing, expanded analytics, reusable atmosphere profiles for different times of day, customer loyalty points, and gamification features that reward appropriate participation. A small

cafe chain could manage several locations through a centralized account while retaining a separate queue and vibe profile for each venue. Additional development could also improve accessibility, reporting, content controls, and integration with commercial playback services.

The broader customer base extends beyond independent cafes. Restaurants, college dining halls, hotel lobbies, fitness studios, barbershops, and other customer-facing businesses could use the same controlled-request model. Venue-management companies and small chains could use centralized administration to supervise multiple locations. Although the Brewpoint Cafe beta group provides a focused prototype test, these broader users illustrate how the real-world product could scale to organizations with different audiences, operating hours, atmosphere requirements, and management structures.

4 Glossary

API (Application Programming Interface): A set of protocols that allows software applications to communicate with each other.

ASCAP (American Society of Composers, Authors and Publishers): A U.S. performing rights organization that licenses music for commercial use.

BMI (Broadcast Music Inc.): A U.S. performing rights organization that manages commercial music licensing on behalf of songwriters and publishers.

Business Control Center: Rythmiq's owner-facing web dashboard for configuring vibe constraints, managing the song queue, and viewing analytics.

Hardware-Agnostic: Software designed to run on existing devices without requiring specific or proprietary hardware.

LAMP: An open-source web development stack consisting of Linux, Apache, MySQL, and PHP.

MFCD (Major Functional Component Diagram): A visual representation of a system's major components and the data flows between them.

QR Code (Quick Response Code): A two-dimensional barcode that can be scanned with a smartphone camera to open a URL or application.

SESAC (Society of European Stage Authors and Composers): A U.S. performing rights organization that provides commercial music licensing.

MusicBrainz: A community-maintained music database planned to provide song metadata for search and request evaluation.

SoundMachine: A third-party commercial music streaming service planned for legal audio playback in Rythmiq venues.

Stripe API: A third-party payment interface planned to process optional pay-to-request transactions.

Vibe Engine: Rythmiq's proprietary validation logic for evaluating customer song requests against venue-configured constraints before queuing them.

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